

Table 37.1 Essential Elements in Plants

Element (Form Primarily Absorbed by Plants)	% Mass in Dry Tissue	Major Function(s)	Early Visual Symptom(s) of Nutrient Deficiencies
Macronutrients			
Carbon (CO ₂)	45%	Major component of organic compounds	Poor growth
Oxygen (CO ₂)	45%	Major component of organic compounds	Poor growth
Hydrogen (H ₂ O)	6%	Major component of organic compounds	Wilting, poor growth
Nitrogen (NO ₃ ⁻ , NH ₄ ⁺)	1.5%	Component of nucleic acids, proteins, and chlorophyll	Chlorosis at tips of older leaves (common in heavily cultivated soils or soils low in organic material)
Potassium (K ⁺)	1.0%	Enzyme cofactor; major solute functioning in water balance; operation of stomata	Mottling of older leaves, drying of leaf edges; weak stems; roots poorly developed (common in acidic or sandy soils)
Calcium (Ca ²⁺)	0.5%	Important component of middle lamella and cell walls; maintains membrane function; signal transduction	Crinkling of young leaves; death of terminal buds (common in acidic or sandy soils)
Magnesium (Mg ²⁺)	0.2%	Component of chlorophyll; cofactor of many enzymes	Chlorosis between veins, found in older leaves (common in acidic or sandy soils)
Phosphorus (H ₂ PO ₄ ⁻ , HPO ₄ ²⁻)	0.2%	Component of nucleic acids, phospholipids, ATP	Healthy appearance but very slow development; thin stems; purpling of veins; poor flowering and fruiting (common in acidic, wet, or cold soils)
Sulfur (SO ₄ ²⁻)	0.1%	Component of proteins	General chlorosis in young leaves (common in sandy or very wet soils)
Micronutrients			
Chlorine (Cl ⁻)	0.01%	Photosynthesis (water-splitting); functions in water balance	Wilting; stubby roots; leaf mottling (uncommon)
Iron (Fe ³⁺ , Fe ²⁺)	0.01%	Respiration; photosynthesis: chlorophyll synthesis; N ₂ fixation	Chlorosis between veins, found in young leaves (common in basic soils)
Manganese (Mn ²⁺)	0.005%	Active in formation of amino acids; activates some enzymes; required for water-splitting step of photosynthesis	Chlorosis between veins, found in young leaves (common in basic soils rich in humus)
Boron (H ₂ BO ₃ ⁻)	0.002%	Cofactor in chlorophyll synthesis; role in cell wall function; pollen tube growth	Death of meristems; thick, leathery, and discolored leaves (occurs in any soil; most common micronutrient deficiency)
Zinc (Zn ²⁺)	0.002%	Active in formation of chlorophyll; cofactor of some enzymes; needed for DNA transcription	Reduced internode length; crinkled leaves (common in some geographic regions)
Copper (Cu ⁺ , Cu ²⁺)	0.001%	Component of many redox and lignin-biosynthetic enzymes	Light green color throughout young leaves, with drying of leaf tips; roots stunted and excessively branched (common in some geographic regions)
Nickel (Ni ²⁺)	0.001%	Nitrogen metabolism	General chlorosis in all leaves; death of leaf tips (common in acidic or sandy soils)
Molybdenum (MoO ₄ ²⁻)	0.0001%	Nitrogen metabolism	Death of root and shoot tips; chlorosis in older leaves (common in acidic soils in some geographic areas)

MAKE CONNECTIONS Explain why CO₂, rather than O₂, is the source of much of the dry mass oxygen in plants. See Concept 10.1.



Mastering Biology

Interview with Gloria Coruzzi:
Assimilating more nitrogen into plants



► **Figure 37.8 The most common mineral deficiencies, as seen in maize leaves.** Mineral deficiency symptoms may vary in different species. In maize, nitrogen deficiency is evident in a yellowing that starts at the tip and moves along the center (midrib) of older leaves. Phosphorus-deficient maize plants have reddish purple margins, particularly in young leaves. Potassium-deficient maize plants exhibit “firing,” or drying, along tips and margins of older leaves.



Healthy



Nitrogen-deficient



Phosphorus-deficient



Potassium-deficient